Something we missed

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\bar{y} = \beta_0 + \beta_1 \bar{x}$$ $(\bar{x}, \bar{y})$ is on regression line

I. Recap

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{\beta_0, \beta_1}{\operatorname{argmin}}\; Q(\beta_0, \beta_1)$$

$$Q(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \quad \text{or}$$

$$= \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

Sum of squared residuals

OR SSE / SSR

Closed form solutions:

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

On HW 1: show that $\hat{\beta}_1 = \dfrac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$

Just show that this is equivalent to the one we derived.

$$= \frac{Cov(x,y)}{Var(x)}$$

II. Properties of Least Squares (LS) estimates

Are these estimates good?

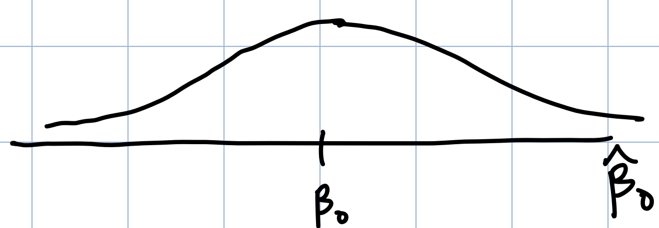
<u>Gauss-Markov Theorem</u>

Under the SLR assumptions, the LS estimates $\hat{\beta}_0$ + $\hat{\beta}_1$ are:

1. Unbiased for β_0 + β_1

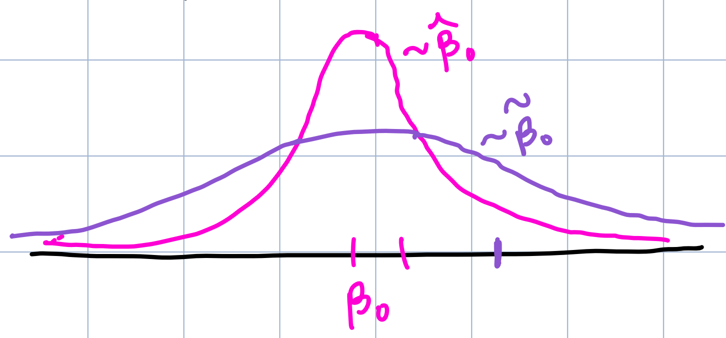
$$E(\hat{\beta}_0) = \beta_0$$

$$E(\hat{\beta}_1) = \beta_1$$



2. The LS estimates are the best linear unbiased estimators (BLUE)

$$Var(\hat{\beta}_0) \leq Var(\tilde{\beta}_0)$$ some other linear unbiased estimate



<u>Let's prove these</u>

1. Let's show $E(\hat{\beta}_1) = \beta_1$

Show $$E(\hat{\beta}_1) = E\left[\frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}\right] = \beta_1$$

Firstly: $\sum_i (x_i - \bar{x})^2$ is called SSX "sum of squares for X"

$\sum_i (x_i - \bar{x})(y_i - \bar{y})$ is called SSXY

Lemma: $SSXY = \sum_i (x_i - \bar{x}) y_i$

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i \left(x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x}\bar{y} \right)$$

$$= \sum_i \left((x_i - \bar{x}) y_i - (x_i - \bar{x}) \bar{y} \right)$$

$$\sum_i (x_i - \bar{x}) y_i - \underbrace{\sum_i (x_i - \bar{x}) \bar{y}}$$

$$\bar{y} \sum_i (x_i - \bar{x}) \qquad \bar{x} = \frac{1}{n} \sum_i x_i$$

$$= \bar{y} \, \cancel{(n\bar{x} - n\bar{x})} \to 0$$

Back to proof:

$$\hat{\beta}_1 = \frac{SSXY}{SSX} = \frac{\sum_i (x_i - \bar{x}) y_i}{SSX} = \sum_i \underbrace{\left(\frac{x_i - \bar{x}}{SSX} \right)}_{k_i} y_i$$

think of this as a constant

$$\hat{\beta}_1 = \sum_i k_i y_i$$

$$E(\hat{\beta}_1) = E\left(\sum_i k_i y_i \right) = \sum_i E(k_i y_i)$$

$$= \sum_i k_i E(y_i)$$

$$= \sum_i k_i \, E\left[\beta_0 + \beta_1 x_i + \varepsilon_i\right]$$

$$= \sum_i k_i \left[E(\beta_0) + E(\beta_1 x_i) + E(\varepsilon_i)\right]$$

$$= \sum_i k_i \left[\beta_0 + \beta_1 x_i + 0\right]$$

$$= \sum_i k_i \beta_0 + \sum_i \beta_1 x_i k_i$$

$$= \underbrace{\beta_0 \sum_i k_i} + \beta_1 \sum_i x_i k_i$$

$$\sum_i k_i = \sum_i \frac{x_i - \bar{x}}{SSX} = \frac{1}{SSX} \sum_i (x_i - \bar{x}) = \frac{1}{SSX}\left[\cancelto{0}{n\bar{x} - n\bar{x}}\right]$$

$$\sum_i k_i x_i = \sum_i \left(\frac{x_i - \bar{x}}{SSX}\right) x_i = \frac{1}{SSX} \sum (x_i - \bar{x}) x_i$$

Use the lemma!

$$\sum_i (x_i - \bar{x}) x_i = \sum_i (x_i - \bar{x})(x_i - \bar{x})$$

$$= \sum_i (x_i - \bar{x})^2$$

$$= SSX$$

$$\Rightarrow \sum_i k_i x_i = \frac{1}{SSX} \cdot SSX = 1$$

$$\Rightarrow \quad \beta_0 \cancelto{0}{\sum_i k_i} + \beta_1 \cancelto{1}{\sum_i k_i X_i}$$

$$= 0 + \beta_1$$

$$= \beta_1$$

You do β_0

2. We want to show

$$Var(\hat{\beta}_1) \leq Var(\tilde{\beta}_1)$$ for some other linear unbiased $\tilde{\beta}_1$

$$Var(\hat{\beta}_1) = Var\left(\sum_i k_i y_i\right)$$

assuming covariance $= 0$

y_i's are uncorrelated since ε_i's are

$$= \sum_i Var(k_i y_i)$$

$$= \sum_i k_i^2 Var(y_i)$$

$$= \sum_i k_i^2 \sigma^2$$

$$= \sigma^2 \sum_i k_i^2$$

$$\sum_i k_i^2 = \sum_i \left(\frac{X_i - \bar{X}}{SSX}\right)^2 = \sum_i \frac{(X_i - \bar{X})^2}{SSX^2}$$

$$= \frac{1}{SSX^2} \sum_i (x_i - \bar{x})^2$$

$$= \frac{1}{SSX^2} SSX$$

$$= \frac{1}{SSX}$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SSX}$$

Can we show $Var(\tilde{\beta}_1) \geq \frac{\sigma^2}{SSX}$?

$$\tilde{\beta}_1 = \sum_i (K_i + d_i) y_i = \sum_i \tilde{K}_i y_i$$

linear combo of y

where $\tilde{K}_i = k_i + d_i$

$d_i \neq 0$

$$Var(\tilde{\beta}_1) = Var\left(\sum_i (k_i + d_i) y_i\right)$$

$$= \sum_i Var\left[(k_i + d_i) y_i\right]$$

$$= \sum_i (k_i + d_i)^2 \, Var\, y_i$$

$$= \sigma^2 \sum_i (k_i + d_i)^2$$

$$= \sigma^2 \sum_i \left(k_i^2 + 2k_i d_i + d_i^2\right)$$

$$= \sigma^2\Big(\sum_i k_i^2 + \underbrace{2\sum_i k_i d_i + \sum_i d_i^2}_{\geq 0}\Big)$$

$$\sum d_i^2 > 0$$

$$2\sum_i k_i d_i = 0$$

$$\Rightarrow \quad \sigma^2\Big(\sum_i k_i^2 + 2\sum_i k_i d_i + \sum_i d_i^2\Big) > \sigma^2 \sum_i k_i^2$$

III Summarize

$$E(\hat{\beta}_1) = \beta_1$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SSX}$$

· What about β_0 ?

Can we show $E(\hat{\beta}_0) = \beta_0$?

$Var(\hat{\beta}_0) \leq Var(\tilde{\beta}_0)$?

$$Var(\hat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{X}^2}{SSX}\right)$$

IV Sampling Distributions of $\hat{\beta}_0$ + $\hat{\beta}_1$

If we assume $\varepsilon_i \sim N(0, \sigma^2)$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{SSX}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{X}^2}{SSX}\right)\right)$$